

Chapter 8: Torque-free Motion & Stability

Dr. Alexander Wittig

Welcome back!

- Any questions on previous weeks content?
- What was last week's content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- Torque-free motion of rotationally symmetric satellite
 - precession rates
 - applications
- Stability of rotations & Damping
 - stable rotation axes
 - damping, effect on stability

Analytical Solutions

Euler's equation in body fixed frame:

$$\vec{T} = \frac{d}{dt}\vec{H} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

constant in body fixed frame!

Expanded in body fixed frame:

$$\begin{pmatrix} T_1 \\ T_2 \\ T_3 \end{pmatrix} = \begin{pmatrix} I_1 \dot{\omega}_1 \\ I_2 \dot{\omega}_2 \\ I_3 \dot{\omega}_3 \end{pmatrix} - \begin{pmatrix} (I_2 - I_3)\omega_2\omega_3 \\ (I_3 - I_1)\omega_1\omega_3 \\ (I_1 - I_2)\omega_1\omega_2 \end{pmatrix}$$

Top (classical mechanics)

Rigid body rotating freely around arbitrary fixed point with torque from a uniform gravitational field.

Only 3 integrable (analytically solvable) tops exist:

Euler

Lagrange

Kovalevskaya



Wikipedia

Career

- Doctorate on PDEs, Dynamics of Saturn rings, elliptic integrals (University of Göttingen, 1874)
- Lecturer then Professor (Stockholm University, 1884/1889)

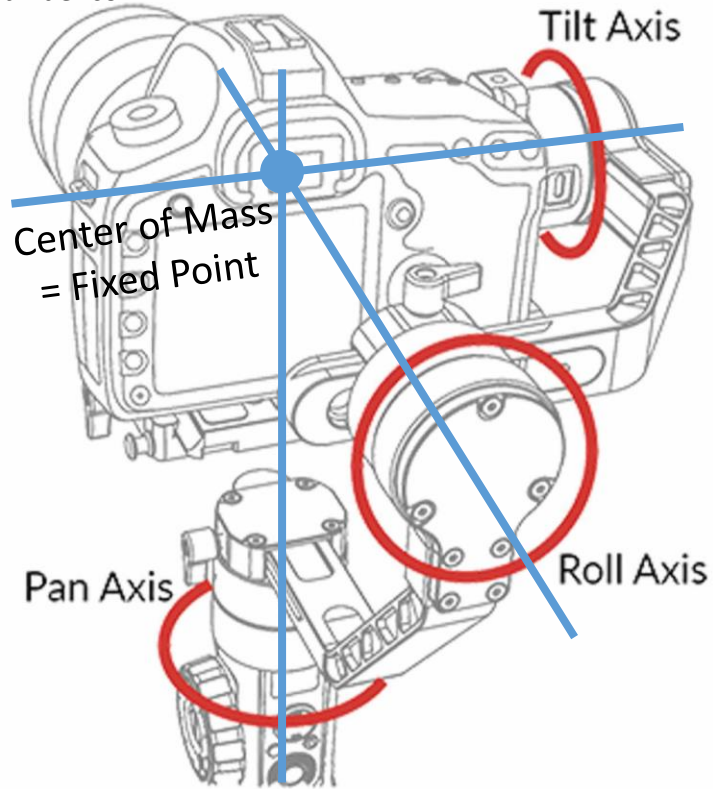
Achievements

- First women to earn (modern) Doctorate in Math
- First women full professor in Northern Europe
- Cauchy-Kovalevsky theorem
- Kovalevsky top

Kowalevski, Sophie, "*Sur le probleme de la rotation d'un corps solide autour d'un point fixe*", Acta Mathematica, **12** (1): 177–232 (1889)

Integrable Tops

studiobinder.com



- Arbitrary body
- Fixed point: CoM
- No torque

Euler

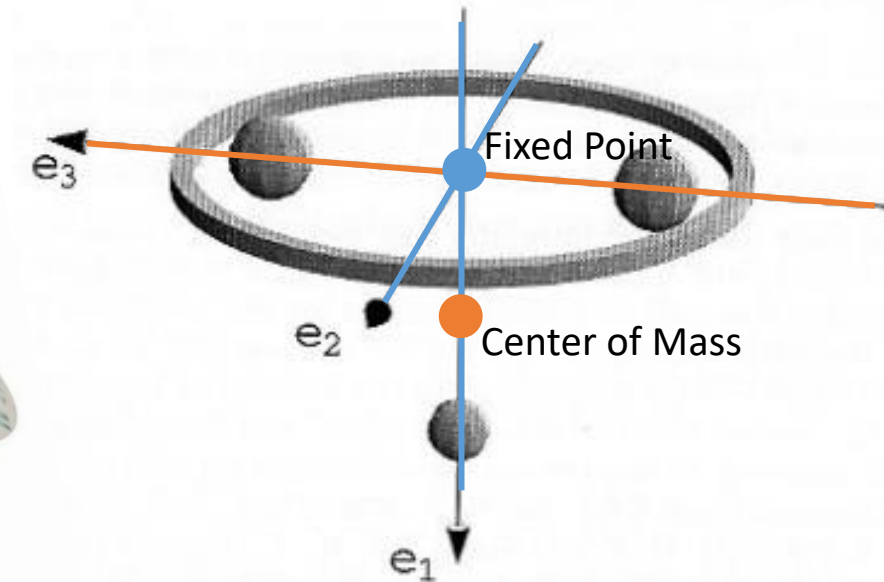
Rex London



- Rotationally symmetric
- Fixed point: on symmetry axis
- Gravity torque

Lagrange

P.H. Richter, H. R. Dullin, A. Wittek
<https://av.tib.eu/media/10361>



- Rotationally symmetric, specific ratio (2:2:1)
- Fixed point: in symmetry plane
- Gravity torque

Kovalevskaya

Torque-Free Motion

Euler & rotational symmetry

Special case: $I_1 = I_2$ (rotationally symmetric *body of rotation*)

$$\begin{pmatrix} I_1 \dot{\omega}_1 \\ I_1 \dot{\omega}_2 \\ I_3 \dot{\omega}_3 \end{pmatrix} = \begin{pmatrix} -(I_3 - I_1) \omega_2 \omega_3 \\ (I_3 - I_1) \omega_1 \omega_3 \\ 0 \end{pmatrix}$$

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$



Body of Rotation: Torque-free Motion

Special case: $I_1 = I_2$ (rotationally symmetric *body of rotation*)

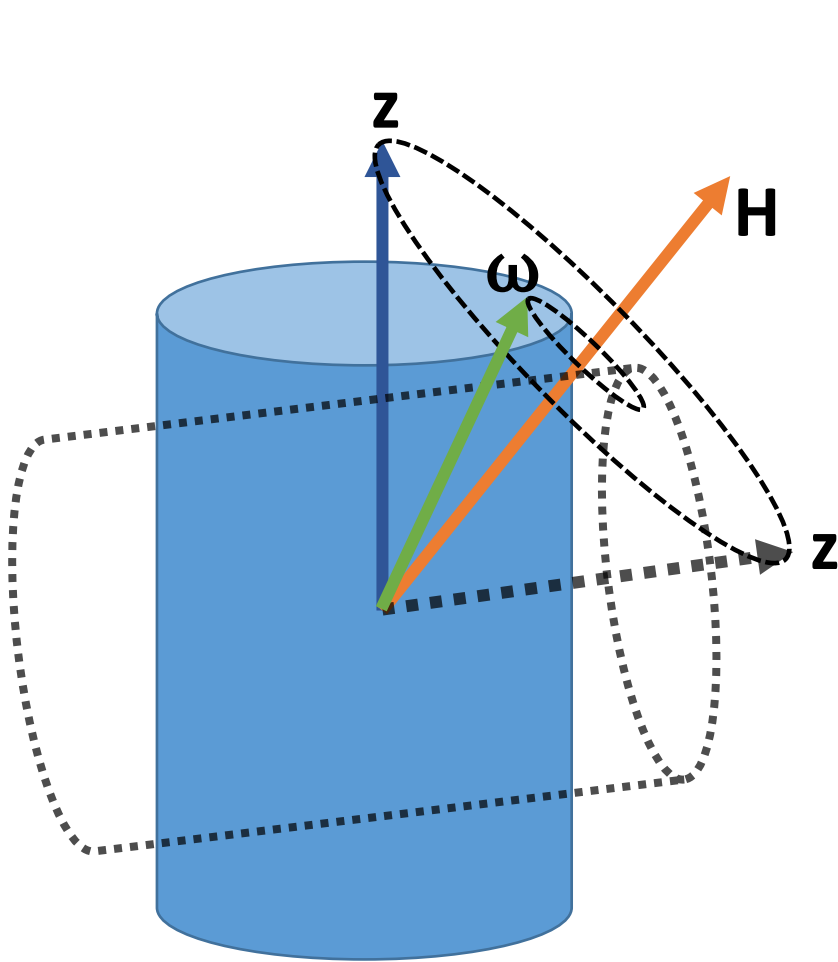
$$\begin{pmatrix} I_1 \dot{\omega}_1 \\ I_1 \dot{\omega}_2 \\ I_3 \dot{\omega}_3 \end{pmatrix} = \begin{pmatrix} -(I_3 - I_1) \omega_3 \\ (I_3 - I_1) \omega_3 \\ 0 \end{pmatrix}$$

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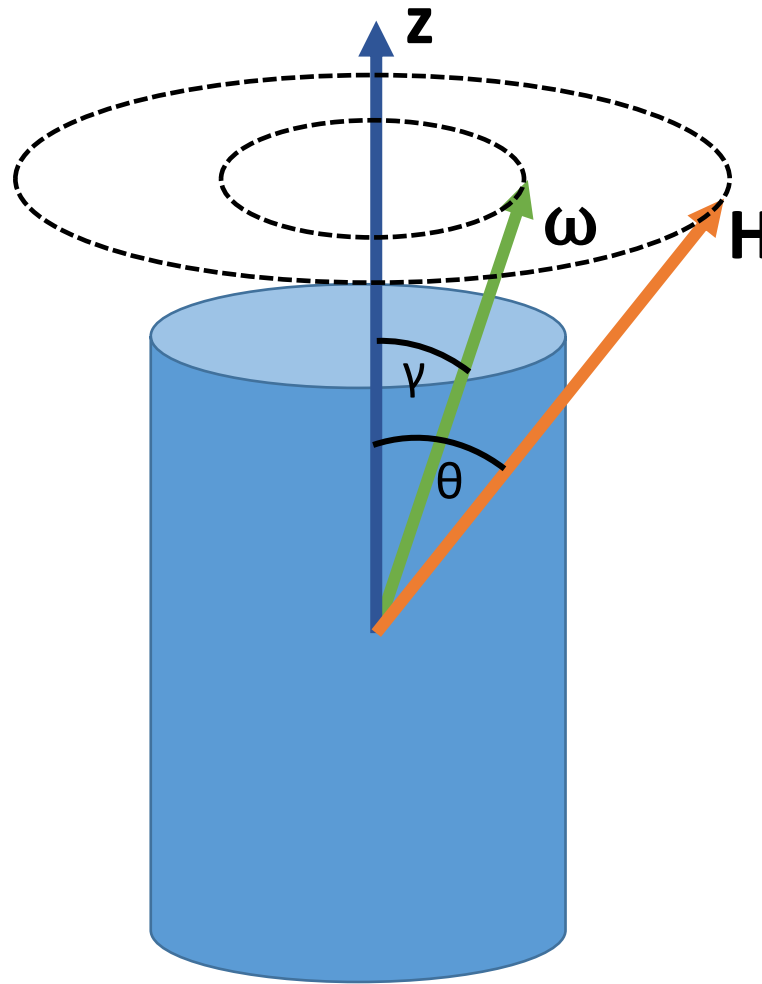


see Chapter 7 Video Lecture 32

Body of Rotation: Torque-free Motion (qualitative)



inertial



body fixed

$$|\vec{H}| = h = \text{const}$$

$$\angle(\vec{H}, \vec{z}) = \theta = \text{const}$$

$$|\vec{\omega}| = \omega = \text{const}$$

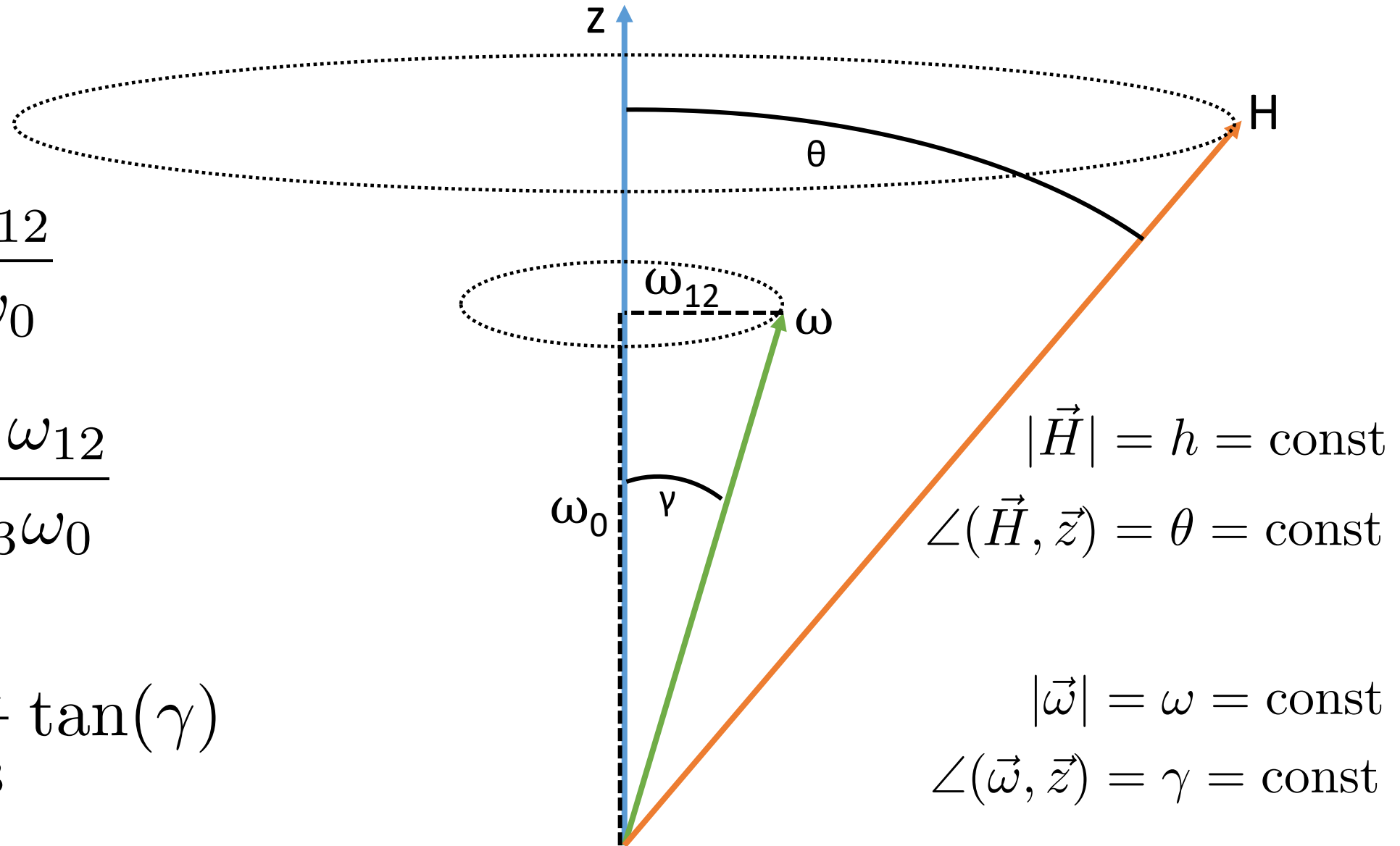
$$\angle(\vec{\omega}, \vec{z}) = \gamma = \text{const}$$

Body of Rotation: Torque-free Motion (geometry)

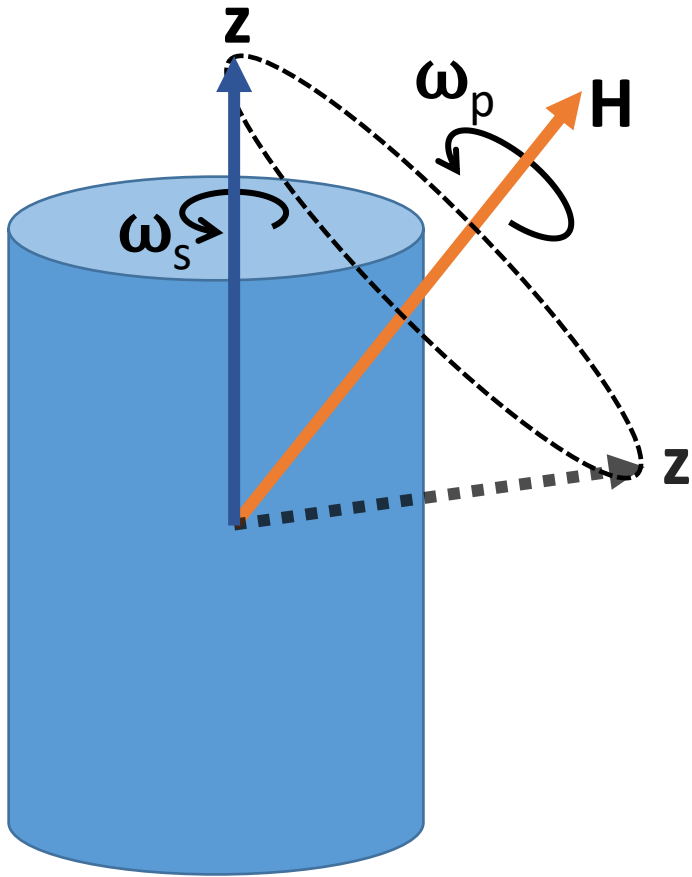
$$\tan(\gamma) = \frac{\omega_{12}}{\omega_0}$$

$$\tan(\theta) = \frac{I_1 \omega_{12}}{I_3 \omega_0}$$

$$\tan(\theta) = \frac{I_1}{I_3} \tan(\gamma)$$



Body of Rotation: Torque-free Motion (quantitative)



inertial

precession
(around fixed
inertial direction $\mathbf{H}$)

spin
(around fixed
body fixed axis $\mathbf{z}$)

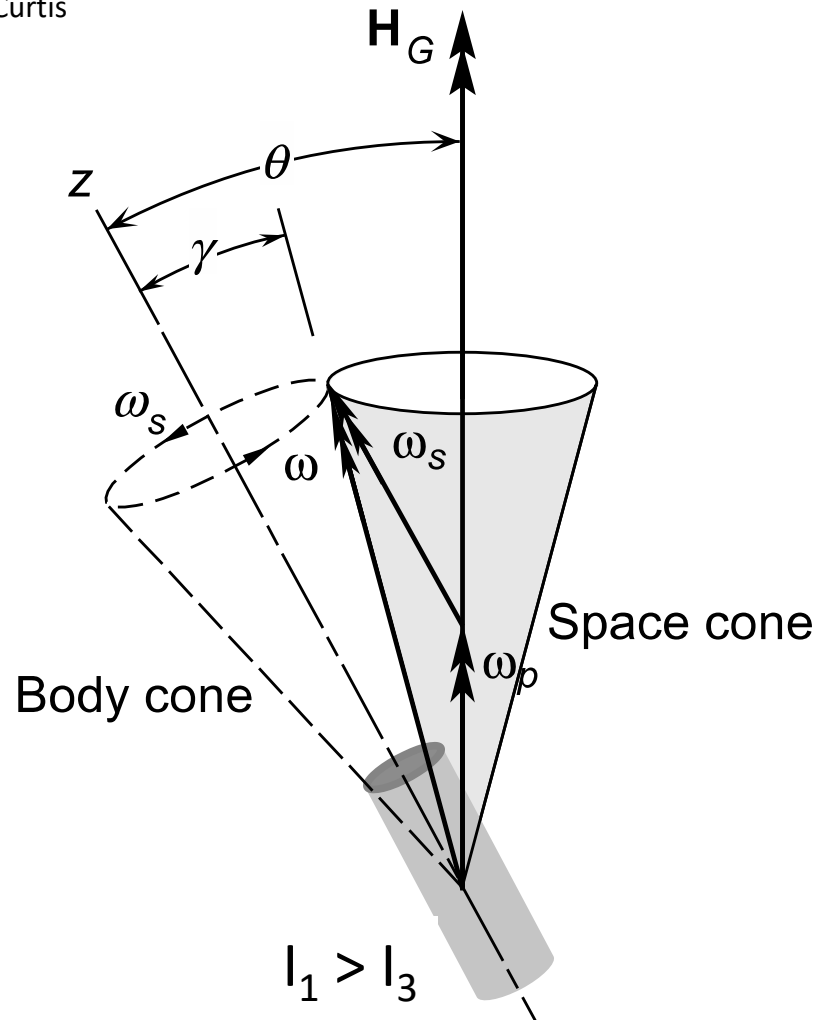
$$\omega_p = \frac{H}{I_1} = \frac{\omega_{12}}{\sin \theta}$$

$$\omega_s = \left(1 - \frac{I_3}{I_1} \right) \omega_0$$

$$= \omega_0 - \frac{\omega_{12}}{\tan \theta}$$

Prolate Bodies

Curtis



(a) Prograde precession
prolate body

Prolate body

- Long and thin
- Symmetry axis has smaller moment of inertia:
 $I_1 > I_3$
- **Minor** axis spinner
- ω_s and ω_p both positive
- **Prograde** precession

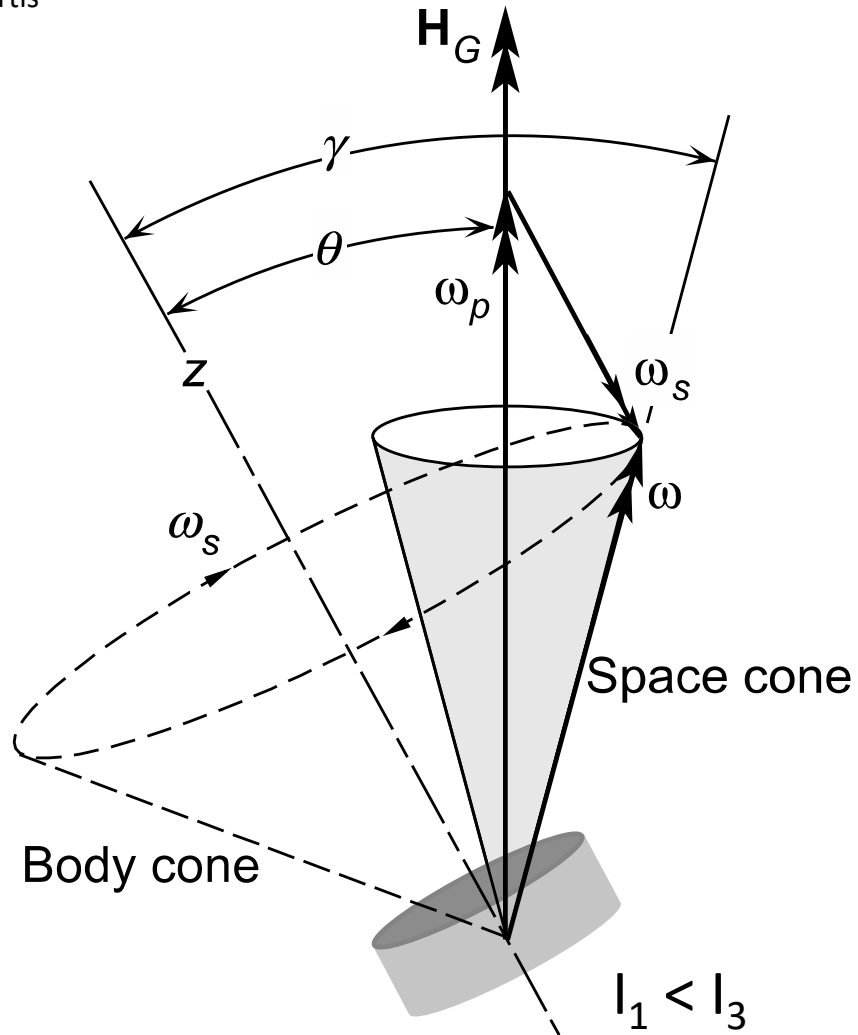
$$\omega_p = \frac{H}{I_1} = \frac{\omega_{12}}{\sin \theta}$$

$$\omega_s = \left(1 - \frac{I_3}{I_1} \right) \omega_0$$

$$= \omega_0 - \frac{\omega_{12}}{\tan \theta}$$

Oblate Bodies

Curtis



(b) Retrograde precession
oblate body

Oblate body

- Flat and wide
- Symmetry axis has larger moment of inertia:
 $I_1 < I_3$
- **Major** axis spinner
- ω_s negative and ω_p positive
- **Retrograde** precession

$$\omega_p = \frac{H}{I_1} = \frac{\omega_{12}}{\sin \theta}$$

$$\omega_s = \left(1 - \frac{I_3}{I_1} \right) \omega_0$$

$$= \omega_0 - \frac{\omega_{12}}{\tan \theta}$$

Spin up entire satellite along principal axis to provide attitude **stabilization**.

Single spin satellite

Pros:

Cons:



Spin up entire satellite along principal axis to provide attitude **stabilization**.

Single spin satellite

Pros:

- protection against perturbations
- continuous sweep
- fully passive

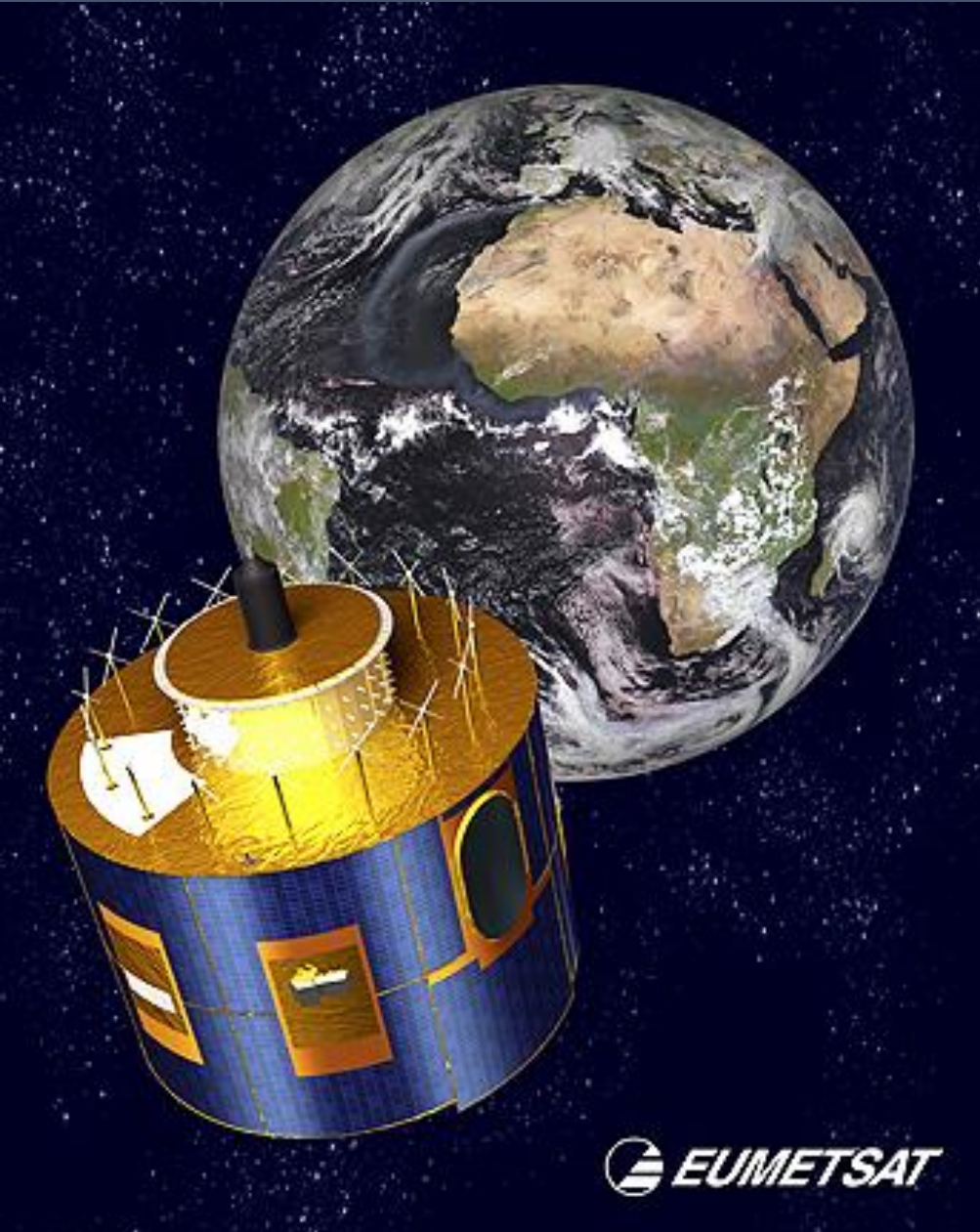
Examples:

- Jupiter upper stages
- Schiaparelli lander
- Pioneer 10 & 11

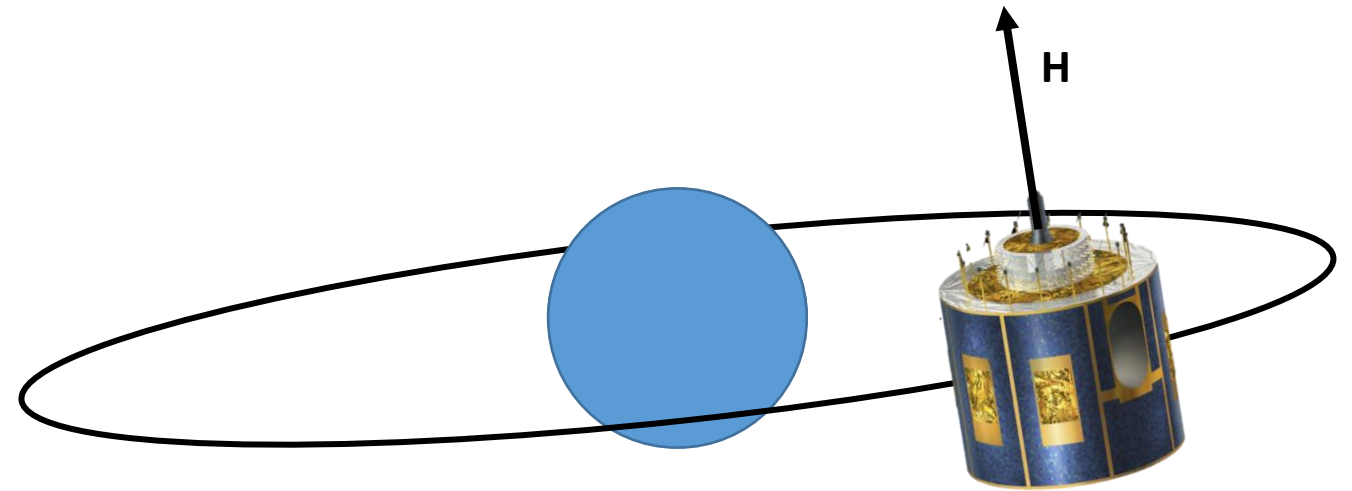
Cons:

- de-spin instruments / antennae
- low maneuverability (see coning maneuver)
- Low solar panel efficiency (only $\frac{1}{2}$ illuminated)





Single spin satellite

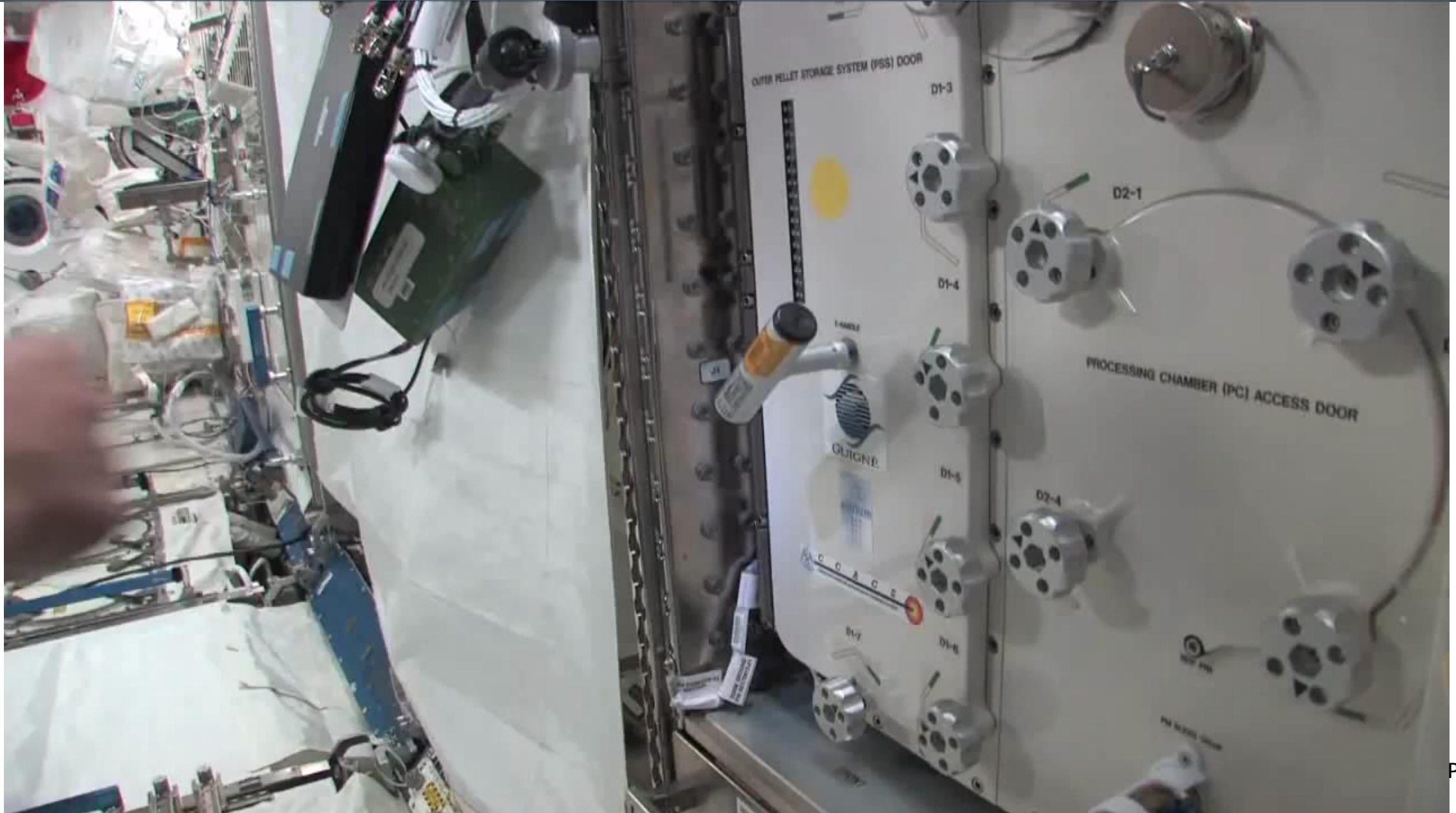


For Earth satellite: angular momentum out of orbital plane

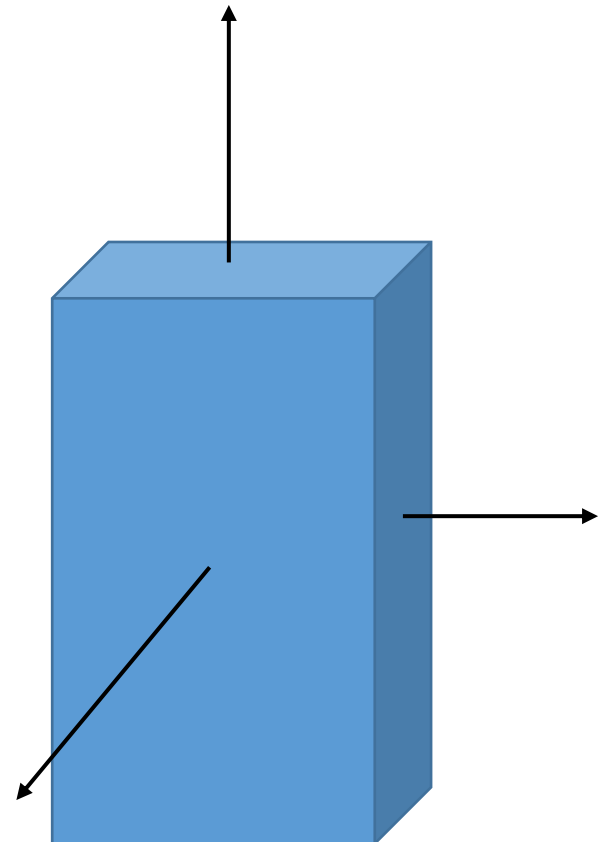
Example: Meteosats

Stability

Arbitrary body

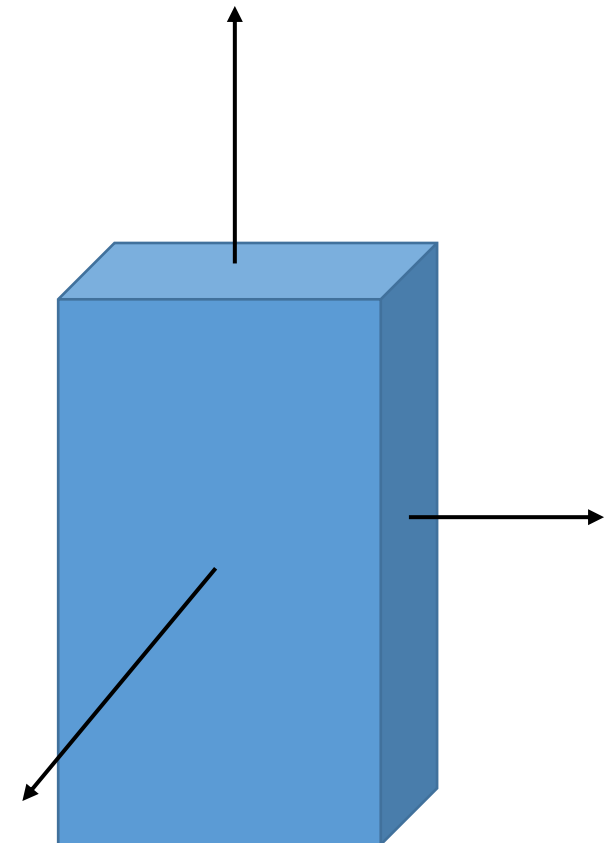


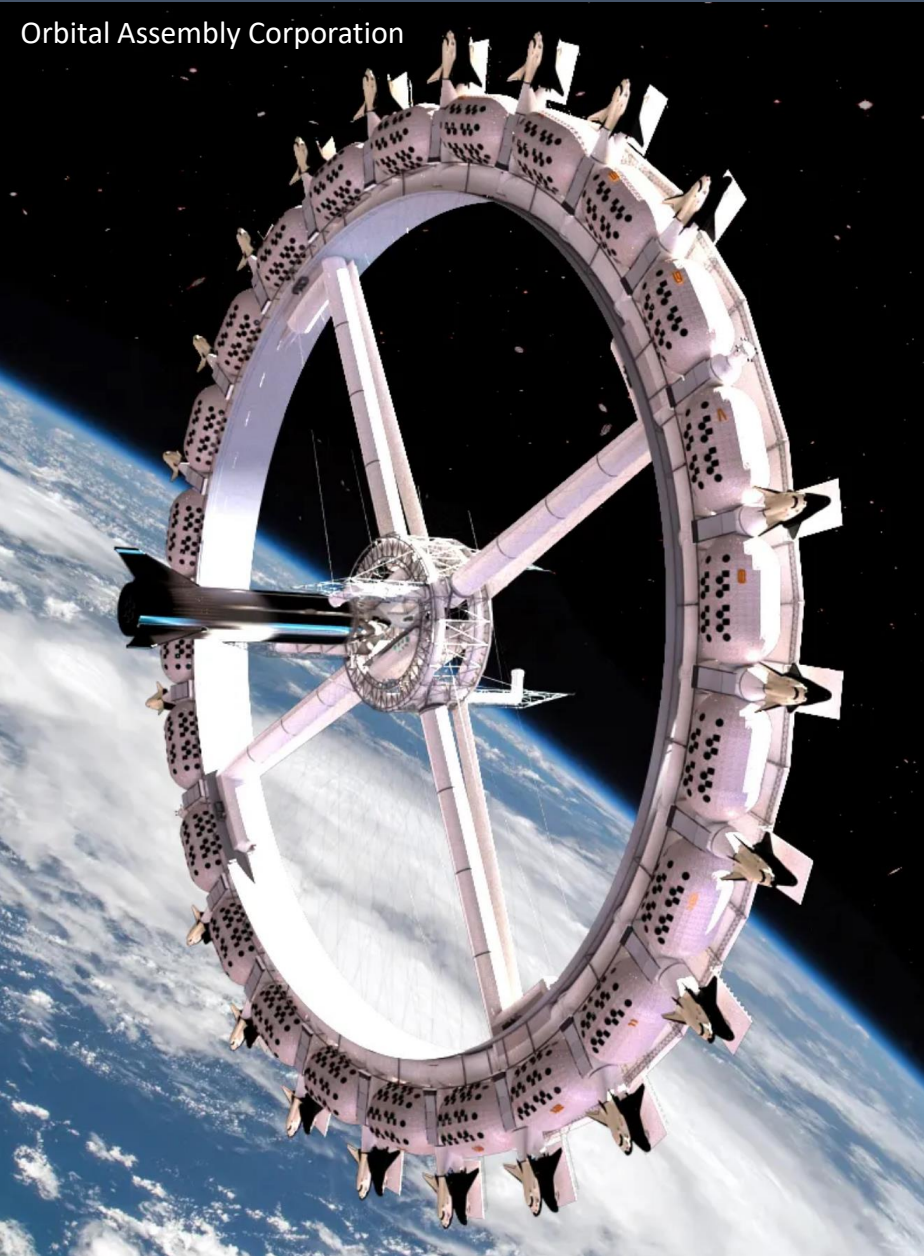
- Torque-free rotations around principal axes (**pure rotations**) are invariant:
 - angular velocity vector remains constant
 - axis of rotation remains constant
 - angular momentum vector remains constant
- Perturbations in real world:



- Torque-free rotations around principal axes (**pure rotations**) are invariant:
 - angular velocity vector remains constant
 - axis of rotation remains constant
 - angular momentum vector remains constant
- Perturbations in real world:
 - one time effects, e.g. micro meteorites, attitude maneuver residuals
 - small time dependent environmental torques e.g. solar radiation pressure
 - internal effects, e.g. thrusting, imperfect alignment

How do rotations react to small perturbations?





Stability: actual rotation axis (angular velocity) stays close to nominal rotation axis over time.

- **Neutral:** actual stays close to nominal
- **Asymptotic:** actual converges to nominal

- Nominal axis could be anything
 - Typically studied around invariant rotations
 - Stability of pure rotations = stability of principal axes

- Different for different types of bodies
 1. Spherically symmetric ($I_1 = I_2 = I_3$)
 2. Body of Rotation ($I_1 = I_2 < I_3$ or $I_1 = I_2 > I_3$)
 3. Arbitrary body ($I_1 < I_2 < I_3$)

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & I_1 \end{pmatrix}$$

Spherically symmetric body: all principal moments equal

Exercise: Perform stability analysis on a spherically symmetric body.

That is, find the axes around which rotations are stable.



LAGEOS (NASA)

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & I_1 \end{pmatrix}$$

Spherically symmetric body: all principal moments equal

Exercise: Perform stability analysis on a spherically symmetric body.

That is, find the axes around which rotations are stable.

Spherically symmetric: every direction is the same

- All axes are principal axes
- All axes are stable

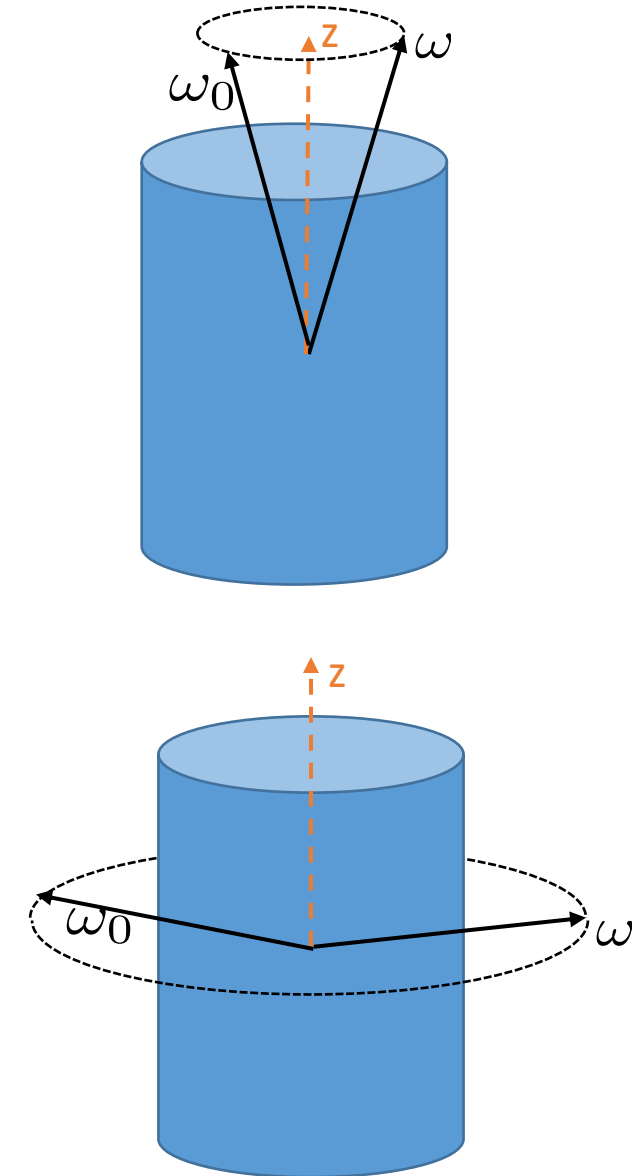


LAGEOS (NASA)

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

Rigid Body of Rotation: two identical principal moments

- Angle of symmetry axis and angular velocity constant
→ symmetry axis: stable
- Transversal axes: not stable
 $\gamma \neq 90^\circ$: precession slowly rotates angular velocity around the cylinder



$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

General Rigid Body: three distinct principal moments

- Linearizing Euler's equations yields stability criteria
- Stability of principal axes:
 - **Largest** and **smallest** principal momenta: **stable**
 - Intermediate principal momentum: **unstable**
- No asymptotic stability without friction/dissipation



$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

General Rigid Body: three distinct principal moments

- Linearizing Euler's equations yields stability criteria
- Stability of principal axes:
 - **Largest and smallest** principal moments of inertia: **stable**
 - Intermediate principal moment: **unstable**
- No asymptotic stability without friction/dissipation

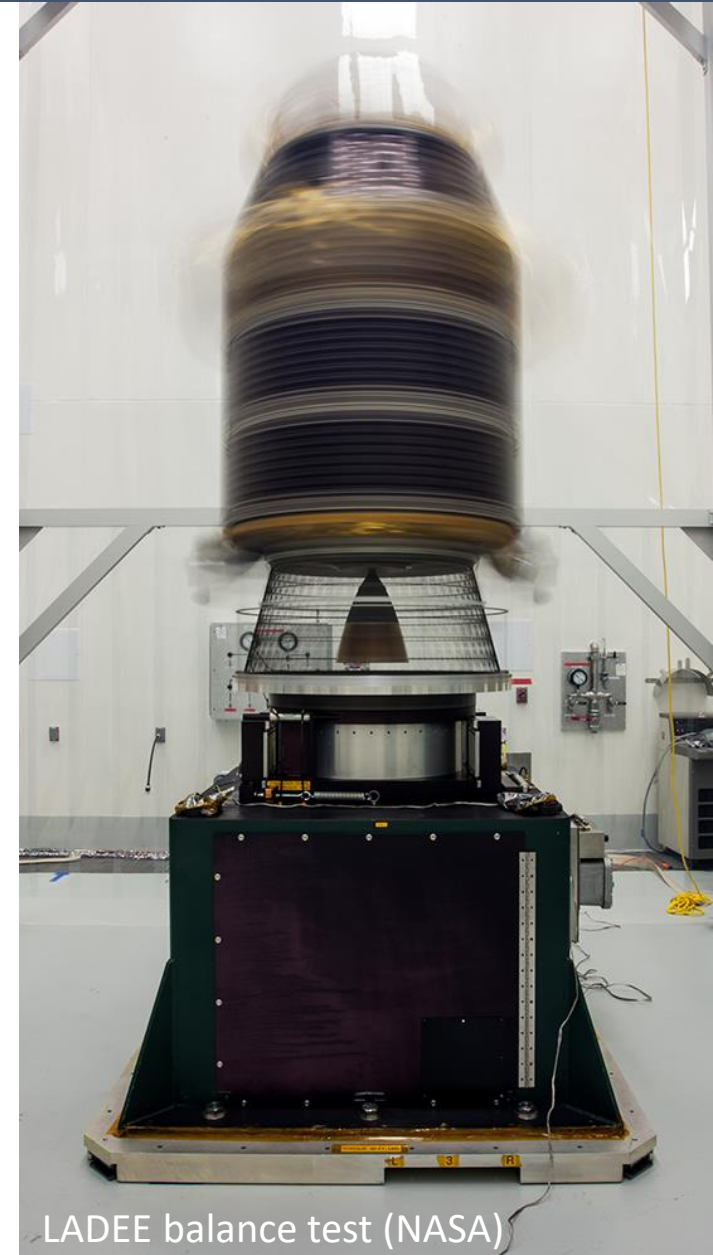
see Chapter 7 Video Lecture 33



Stability

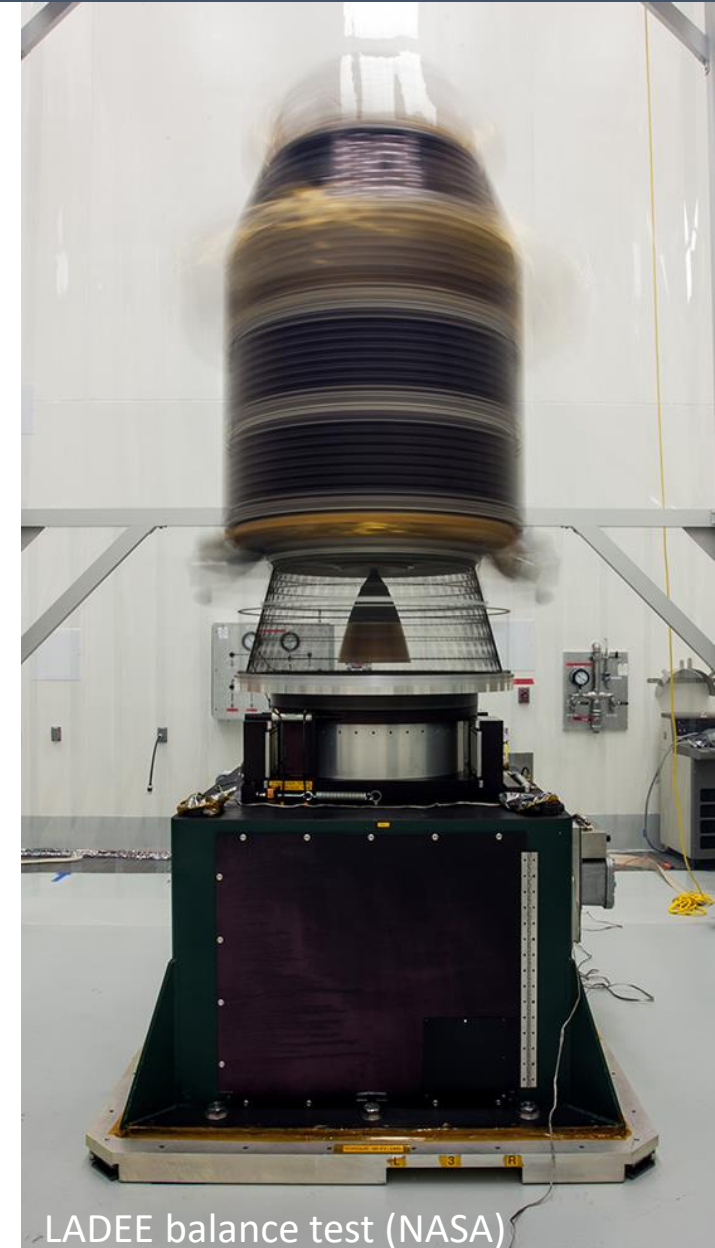
Damping

- Perfectly Rigid Body motion: energy conserved
- Real world bodies dissipate energy:



LADEE balance test (NASA)

- Perfectly Rigid Body motion: energy conserved
 - Real world bodies dissipate energy:
 - Flexing in structure
 - Sloshing of liquids
 - Magnetic hysteresis
 - Intentional damping devices
 - Mathematically:
rotational energy $\dot{T} \leq 0$
- ➔ effect on rotations?



LADEE balance test (NASA)

Arbitrary body

- Assuming small deformations
 - no change in inertia matrix
 - no external torques
 - angular momentum magnitude conserved
- Only the **largest** principal moment is **asymptotically stable!**
- Intuitively: System converges towards lowest energy state

Arbitrary body

- Assuming small deformations
 - no change in inertia matrix
 - no external torques
 - angular momentum magnitude conserved
- Only the **largest** principal moment is **asymptotically stable**!
- Intuitively: System converges towards lowest energy state

Consequences for rotation-symmetric body

Only **oblate** spinners are **asymptotically stable**:

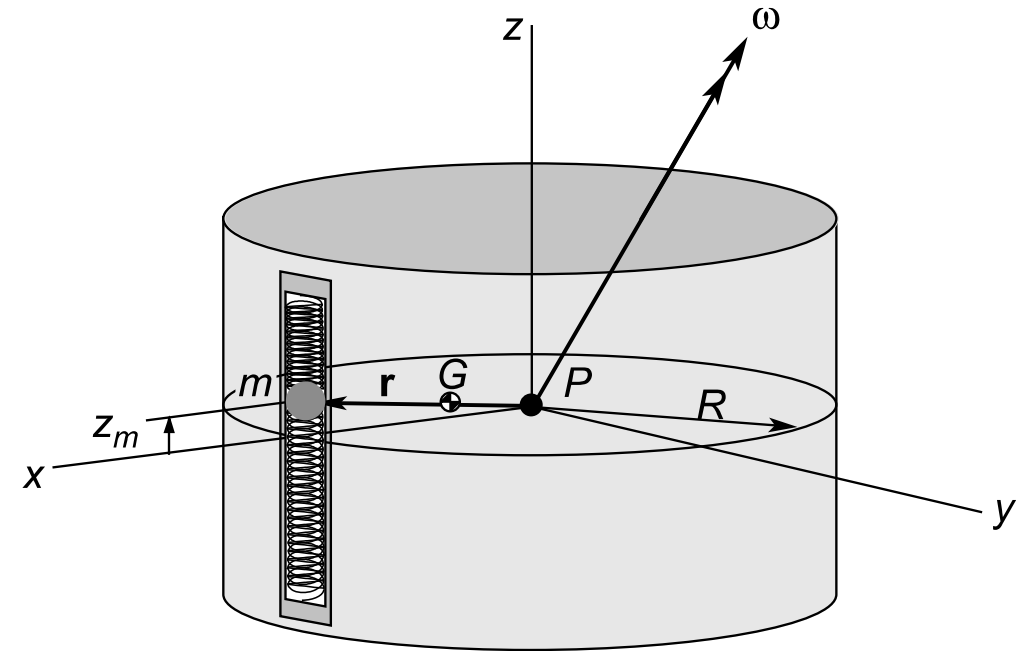
- Any ω_{12} will dissipate
- $|\vec{H}|$ remains constant, but orientation aligns with z
- Only rotation around body symmetry axis (z) remains
- z-axis is **asymptotically stable**

Idea: use intentional dissipation to reach asymptotically stable state

Nutation dampers dissipate energy through heat

- spring/mass (mechanical)
- liquid/mass (mechanical)
- hysteresis (magnetic)

Passive stabilization against small perturbations
Common on CubeSats (hysteresis rods)



Nutation damper schematic (Curtis)

Nutation Damper

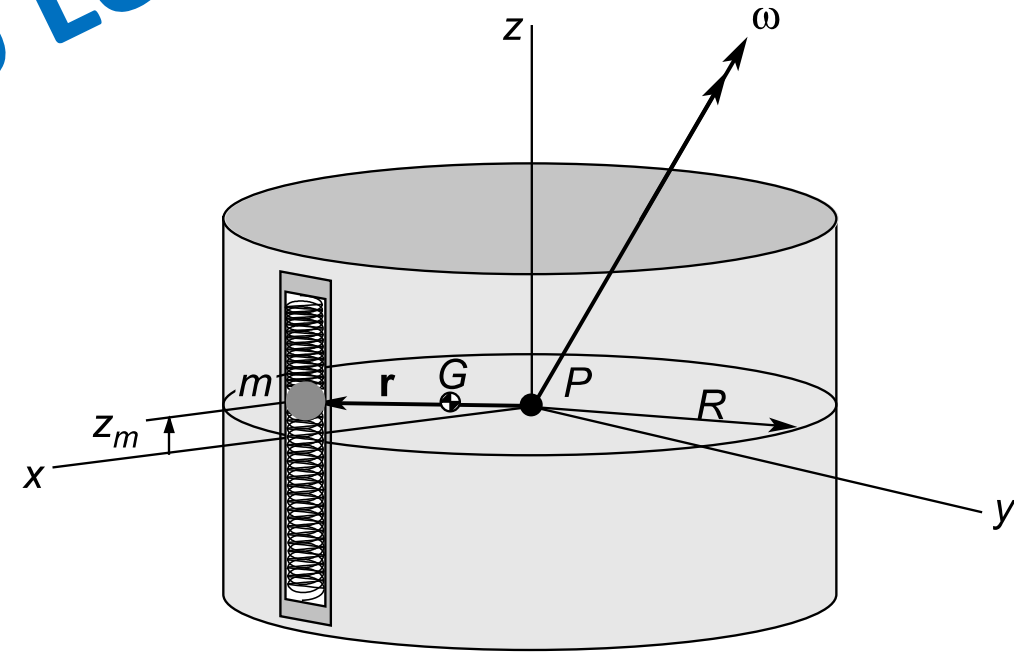
Idea: use intentional dissipation to reach asymptotically stable $s^+ = 0$

Nutation dampers dissipate energy through heat

- spring/mass (mechanical)
- liquid/mass (mechanical)
- hysteresis (magnetic)

Passive stability

Common configurations (hysteresis rods)



Nutation damper schematic (Curtis)

see Chapter 7 Video Lecture 34

- Torque free motion
- body of rotation
- Prolate, oblate body
- Prograde, retrograde rotation
- angular velocity, angular momentum cones
- body spin around symmetry axis
- precession around angular momentum axis
- spin and precession rates constant
- angles between body z-axis, angular velocity and angular momentum constant
- Spin-stabilization
- Stability
 - neutral
 - asymptotic
- Stable/Unstable spin axes
- Dissipation
- Nutation damping

Chapter 8: Torque-free Motion & Stability

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